Experiment 8

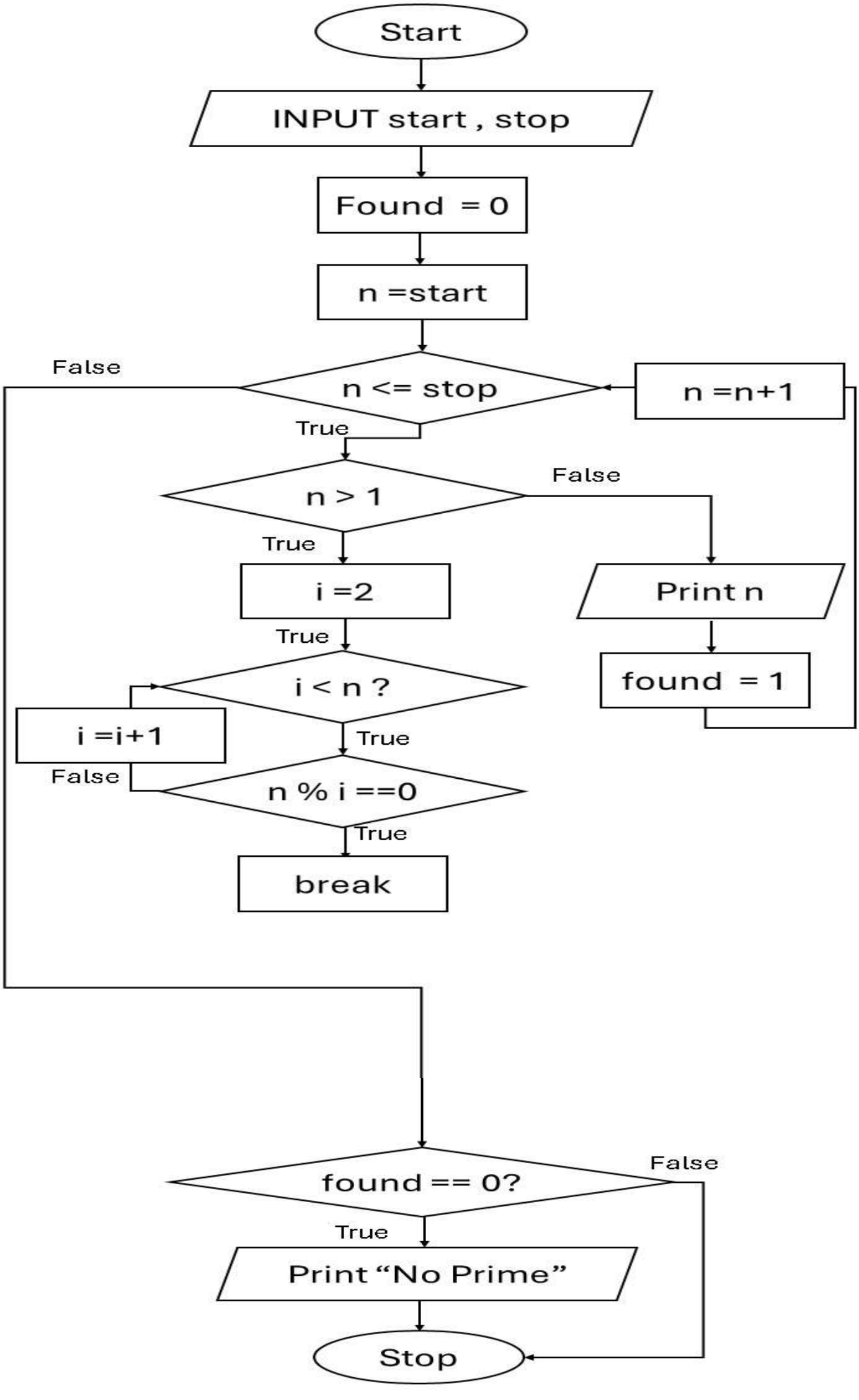
8.1.2 Prime Numbers in a

Algorithm :

```
Step 1: Start
Step 2: Input Start , Stop
Step 3: found = 0
Step 4: Set for loop n in range ( start , stop+1)
        if n > 1:
                Set for loop I in range (2, n):
                        if n % i==0:
                                then n is not prime
                        break the loop
                else :
                        n is prime
                        found =1
                        print  n

Step 5 : If found == 0 :
        i.e n is not prime :
        print not prime
Step 6: Stop
```

Flowchart :


Start
INPUT start , stop
Found = 0
n =start
n <= stop
False
True
n =n+1
n > 1
False
True
i =2
Print n
True
found = 1
i < n ?
i =i+1
True
False
n % i ==0
True
break
found == 0?
False
True
Print “No Prime”
Stop

8.1.2. Prime Numbers in a Given Range

09:42

Write a Python program to print all prime numbers within a given range (inclusive of both start and end values).

Input Format:

- First line contains an integer representing the start of the range.
- Second line contains an integer representing the end of the range.

Output Format:

- Print all prime numbers in the given range (including start and end if they are prime). Print each prime number on a separate line.
- If no prime numbers exist in the range, print: "No primes".

Sample Test Cases +

Explorer

primeNu...

Submit

```python
start = int(input())
stop = int(input())

f = False

for num in range(start, stop + 1):
    if num > 1:
        flag = True
        for i in range(2, int(num ** 0.5) + 1):
            if num % i == 0:
                flag = False
                break
        if flag:
            print(num)
            f = True
if not f:
    print("No primes")
```

Debugger

Average time	Maximum time
0.006 s	0.008 s
6.44 ms	8.00 ms

5 out of 5 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 7 ms Debug

Expected output	Actual output
10	10
20	20
11	11
13	13
17	17
19	19

Terminal Test cases